



Experimental and kinetic modeling study of *n*-propanol and *i*-propanol combustion: Flow reactor pyrolysis and laminar flame propagation

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ABSTRACT

Flow reactor pyrolysis and laminar flame propagation are investigated for *n*-propanol and *i*-propanol, which are the smallest alcohol fuels with isomeric structures. Pyrolysis products of propanol isomers at 0.04–1 atm are detected using the synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS). Experimental observations demonstrate ethylene and propene are respective dominant hydrocarbon products in the *n*-propanol and *i*-propanol pyrolysis, while the most abundant oxygenated products are formaldehyde, acetaldehyde and ethenol in the *n*-propanol pyrolysis and acetone and acetaldehyde in the *i*-propanol pyrolysis. Higher concentrations of aromatic and oxygenated pollutants are observed in the *i*-propanol pyrolysis. The laminar burning velocities of propanol isomers are measured in a high-pressure constant-volume cylindrical combustion vessel at the initial temperature of 423 K, pressures of 1–10 atm and equivalence ratios of 0.6–1.5. A general trend that *n*-propanol has much faster LVBs than *i*-propanol is noticed under all investigated conditions. A kinetic model of propanol isomers is developed and validated against the present experimental data, as well as other experimental data in literature covering wide ranges of temperatures, pressures and equivalence ratios. Rate of production analysis and sensitivity analysis are performed together to provide insight into the fuel isomeric effects on key reaction pathways and fuel reactivities. In the flow reactor pyrolysis, different dominant primary decomposition pathways of *n*-propanol and *i*-propanol lead to the differences in both molecular structures and concentration levels of pyrolysis products. In laminar flame propagation, different radical pool distributions of propanol isomers are illustrated and found to be largely influenced by fuel isomeric structures. The linear structure of *n*-propanol promotes the formation of more active radicals like formyl, vinyl and ethyl, while the branched structure of *i*-propanol facilitates the production of more stable radicals including methyl and allyl. Thus, the promoted chain-branching in *n*-propanol flames and enhanced chain-termination in *i*-propanol flames can explain the higher laminar burning velocities and reactivities of *n*-propanol than that of *i*-propanol.

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1. Introduction

As one of the most important biofuels, bio-alcohol has attracted great attentions from the combustion community due to its significant role as renewable engine fuels or fuel additives [1–5]. There is a growing interest in large alcohols with more than two carbon atoms like propanol, butanol and pentanol, due to a number of advantages with respect to those widely used small alcohols like methanol and ethanol, such as high lower heating values (LHV), favorable boiling points, pertinent miscibility with

conventional fuels and unproblematic use in today's engines [1,6]. Large alcohols are full of isomeric structures, which is their most important structural feature compared with methanol and ethanol. Propanol is the smallest alcohol with isomeric structures, including *n*-propanol and *i*-propanol. Recent studies [7–9] showed that propanol isomers can be produced from biomass feedstocks via *Clostridium* species and *Escherichia coli*, while the latter can achieve an industrial production level. Feasibility of using propanol isomers as fuel and fuel additives has also been extensively tested in various engines, including spark ignition [6,10–12], compression ignition [13–16] and homogeneous charge compression ignition [17–19] engines. Though the performance varies at different operations when blended with propanol isomers, a general trend of improved engine performance and reduced emissions can

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be found with propanol addition. Besides the role as biofuels, propanol isomers can also serve as model fuels for larger alcohols with isomeric structures, especially for investigating the influence of different positions of OH moiety on carbon skeleton.

Sarathy et al. [2] reviewed the progress on the experimental studies of propanol isomers until 2014, which was followed by the review of previous experiment studies by Jouzdani et al. in 2017 [20]. Compared to methanol and ethanol, there are only a limited number of experimental investigations on combustion chemistry of propanol isomers. Among these work, many were focused on the measurements of global combustion parameters like ignition delay times [20–24] and laminar burning velocities (LBVs) [25–32]. Among previous LBV investigations of propanol isomers, measurements at high pressures are still limited, especially those performed for both isomers under comparable conditions. Beeckmann et al. [26] performed the LBV measurements of *n*-propanol in a constant-volume spherical combustion vessel at the initial temperature of 373 K and pressure of 10 bar. LBVs of *n*-propanol/air and *i*-propanol/air mixtures at 423 K and 1–10 atm were measured by Galmiche et al. [28] and Togbé et al. [30] in a same combustion vessel, respectively. However, the measurements at 5 and 10 atm were only performed under the stoichiometric or nearly stoichiometric conditions.

Speciation work of propanol isomers were mainly conducted in batch reactor [33–35], flow reactor [36,37], jet-stirred reactor [28,30], shock tube [20], low-pressure laminar premixed flames [38,39] and atmospheric counterflow flames [40,41]. As indicated by the sensitivity analysis [42], fuel-specific chain-initiation reactions generally show great influence on the fuel reactivity during pyrolysis, demonstrating the necessity of pyrolysis investigations on propanol isomers. Among pyrolysis studies of propanol isomers, Barnard and Hughes [33] and Barnard [34] investigated the pyrolysis of *n*-propanol and *i*-propanol in a static reactor at pressures of 15–300 Torr and 20–300 Torr, respectively, while Trenwith [35] studied the *i*-propanol pyrolysis under pressures ranging from 10 to 100 Torr. Stable pyrolysis products of propanol isomers were detected by titrimetric method or gas chromatography (GC). Heyne et al. [36] investigated the decomposition of *i*-propanol in a Variable Pressure Flow Reactor (VPFR) at 12.5 atm and 976–1000 K using GC. Recently, Jouzdani et al. [20] performed the pyrolysis experiments of propanol isomers in a shock tube with the quantitative detection of CO at temperatures of 1150–1550 K and pressures of 3.5–11 atm using quantum cascade laser absorption spectroscopy. It is recognized that, among previous pyrolysis studies, only limited speciation information was provided, while few efforts have been made on exploring fuel isomeric effects under comparable conditions.

Based on the experimental progresses, kinetic models of propanol isomers have been developed [2,21,23,27,28,30,41]. Johnson et al. [23] firstly constructed the kinetic model of propanol isomers by analogy to alkanes and other alcohols in order to interpret the measured ignition delay time data. Subsequently, several kinetic models of propanol isomers were developed by different groups, such as Veloo and Egolfopoulos [27], Togbé et al. [30], Man et al. [21] and Sarathy et al. [2], based on the model of Johnson et al. [23] with some reactions added and updated. These models were mainly validated against speciation data in jet-stirred reactor oxidation and counterflow flames and global combustion parameters including ignition delay times and LBVs.

In this work, the different pyrolysis behaviors of propanol isomers in a flow reactor are investigated at 0.04–1 atm and 904–1333 K, while the LBVs of propanol isomers are measured at 423 K from atmospheric pressure to high pressure (10 atm), providing new data to explore the fuel isomeric effects and validate propanol models. A kinetic model of propanol isomers is developed and validated against a wide range of experimental data, including

Table 1

Experimental conditions of the flow reactor pyrolysis in this work.

P/atm	T/K	Total flow rate/SCCM	Fuel%	Ar%
0.04	990–1333	1000	3	97
0.2	904–1210	1000	3	97
1	904–1133	1000	3	97

both the new data in this work and the literature data, such as the speciation data in shock tube pyrolysis, jet-stirred reactor and flow reactor oxidation, laminar premixed flame and counterflow diffusion flame, as well as global combustion parameters including ignition delay times and LBVs. Modeling analysis is also performed to reveal the crucial reactions responsible for the different combustion behaviors between propanol isomers.

2. Experimental methods

2.1. Flow reactor pyrolysis

The flow reactor pyrolysis experiments of *n*-propanol and *i*-propanol are carried out at the National Synchrotron Radiation Laboratory. Detailed descriptions of the beamline and flow reactor apparatus can be found in our previous work [43–45]. Liquid *n*-propanol (99% purity) and *i*-propanol (99.8% purity) used in both present flow reactor pyrolysis and LBV measurements are provided by Aladdin Shanghai Biochemical Technology Co., Ltd. The experimental conditions of the flow reactor pyrolysis are shown in Table 1. After vaporization, the fuel/Ar mixture (3%/97% in mole) with a total flow rate of 1000 standard cubic centimeters per minute (SCCM) is fed into a flow tube with 224 mm length. The flow tube is made of α -alumina to reduce wall catalytic effect [46–48] and has a 150 mm heating length. A small inner diameter (7 mm) is used to ensure strong radial diffusion effects, reduce radial concentration gradients and achieve adequately homogeneous reaction circumstances according to the experiences in previous laminar flow reactor experiments [49–51]. Temperature profiles along the flow tube are measured using an S-type thermocouple and served as input parameters in the simulations, while the maximum value ($T_{\max}$) is used to denote each condition. Detailed description of the temperature measurement method can be found in our previous work [42]. During the experiment, the pyrolysis species are sampled by a quartz nozzle. The formed molecular beam passes through a nickel skimmer and is then ionized by the tunable synchrotron VUV light. Eventually, a home-made reflectron time-of-flight mass spectrometer (RTOF-MS) is used to detect these ions. Methodologies for the intermediate identification through measurements of photoionization efficiency (PIE) spectra and the mole fraction evaluation were also reported in detail previously [52,53]. The uncertainties of evaluated mole fractions are estimated to be within $\pm 25\%$ for pyrolysis products with known photoionization cross sections (PICs), and a factor of 2 for those with estimated PICs. The PICs used to evaluate the mole fractions of pyrolysis species are available in the online database [54] and the specific PICs reference are provided in Table S2 in the Supplementary materials.

2.2. Laminar burning velocity

The LBV measurements of propanol isomers are performed in a high-pressure constant-volume cylindrical combustion vessel in Shanghai Jiao Tong University. The apparatus has been introduced in detail elsewhere [44, 55] and only brief descriptions will be provided herein. Combustible mixtures of fuel (*n*-propanol or *i*-propanol) and air (79%N₂/21%O₂) are prepared in the premixing

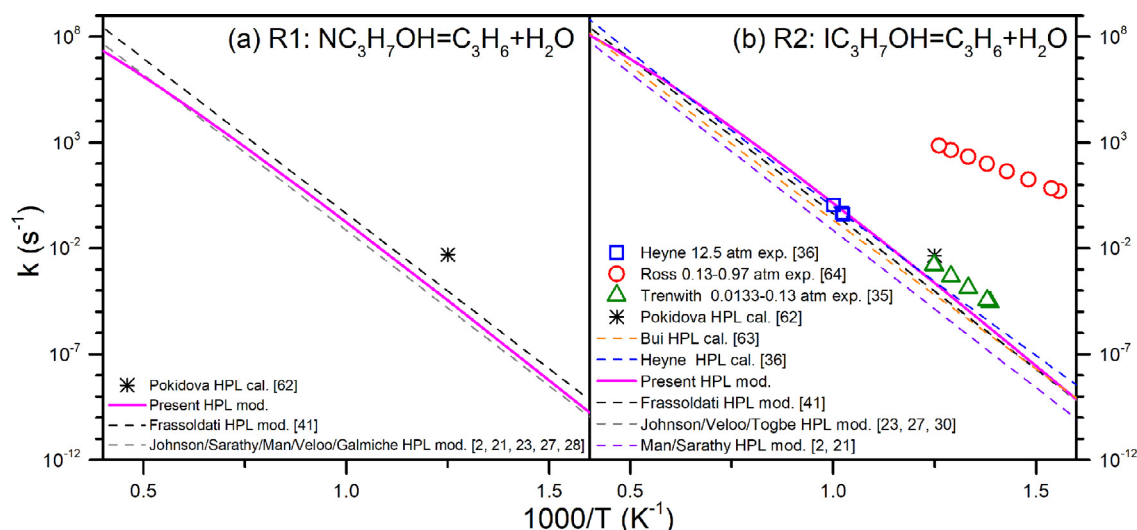


Fig. 1. Comparison of rate constants of (a) R1 and (b) R2 from different sources [2,21,23,27,28,30,35,36,41,62–64]. The abbreviations “exp.”, “cal.” and “mod.” denote rate constants from experiments, calculations and kinetic models, respectively. Solid lines denote the HPL rate constants used in the present model.

vessel by using partial pressure method, which can provide all combustible mixtures needed for a given equivalence ratio at different pressures. During each experiment, the combustible mixture is fed into the combustion vessel and ignited by two horizontally-installed tungsten electrodes. Thus an outwardly propagating spherical flame is formed and recorded simultaneously by a high speed camera (Phantom V310) via a schlieren system which is built up through two 75-mm-diameter apertures in the combustion vessel. The camera is operated at 12,001 frames/sec with a resolution of 480×480 pixels. Temperatures of the premixing vessel, vaporizer, pipelines and combustion vessel are all kept at 423 K. For both *n*-propanol and *i*-propanol, the experiments are performed at the initial pressures of 1–10 atm and equivalence ratios of 0.6 to 1.5, while each condition is repeated for three times. The nonlinear extrapolation method developed by Kelley and Law [56] is adopted to process the recorded flame images and obtain the LBV and Markstein length results. The flame radius range is carefully selected to eliminate the ignition and confinement effects [57–60], mainly 10–23 mm in this work, while the uncertainties of LBVs at different pressures and equivalence ratios are also evaluated using the methods outlined by Moffat [61]. Details of the uncertainty evaluation methods can be found in our previous work [55].

3. Kinetic modeling

The present kinetic model of *n*-propanol ($\text{NC}_3\text{H}_7\text{OH}$) and *i*-propanol ($\text{IC}_3\text{H}_7\text{OH}$) is developed based on our recently reported model of butane isomers [44]. The development of *n*-propanol and *i*-propanol sub-mechanism will be introduced in detail below. As concluded by Sarathy et al. [2], theoretical calculations and experimental measurements for the rate constants of reactions involved in the sub-mechanism of propanol isomers are extremely deficient compared to those of ethanol and butanol isomers. Thus for the propanol models [2,21,23,27,28,30,41], the rate constants need to be referred to analogous reactions of smaller or larger alcohols. In this work, same strategy is adopted for the rate constants of reactions involved in sub-mechanism of propanol isomers. Thermodynamic and transport data of species involved in the sub-mechanism of propanol isomers are taken from Johnson et al. [23] and Man et al. [21]. The reaction mechanism, thermodynamic data and transport data can be found in the *Supplementary materials*.

3.1. Unimolecular decomposition reactions

For the unimolecular decomposition reactions of propanol isomers, two reaction classes are considered in this model, i.e. H_2O elimination (R1 and R2) and bond dissociation reactions. The former reaction class proceeds via the simultaneous stretching and breaking of C–O and C–H bond, leading to the formation of corresponding alkene and H_2O through a four-membered ring transition state. Figure 1 presents the rate constants of water elimination reactions investigated or used in literature [2,21,23,27,28,30,35,36,41,62–64]. It is observed that very few experimental and theoretical work were performed for R1, while large discrepancies exist among the available rate constants of R2. Pokidova et al. [62] calculated the high-pressure limit (HPL) rate constants of R1 and R2 using method of crossing parabolas (MCP) only at 800 K. Bui et al. [63] performed theoretical calculations with variational RRKM method for unimolecular decomposition reactions of *i*-propanol. As displayed in Fig. 1(b), their calculated rate constant of R2 shows large discrepancy with the recently experimental and calculated results of Heyne et al. [36]. Furthermore, the measured rate constants of R2 at low to atmospheric pressures by Ross and Stimson [64] in 1960 and Trenwith [35] in 1975 are much higher than both the high-pressure rate constants measured by Heyne et al. [36] in 2015 and HPL values in all calculation work and kinetic models. For the C–C bond dissociation reactions, among the three reactions (R3–R5), only R5 was calculated by Bui et al. [63], however their calculated results also show large discrepancies with the measurements of Heyne et al. [36].



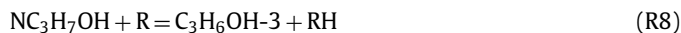
Considering above issues, most unimolecular decomposition reactions of propanol isomers in the present model follow the strategy of previous propanol models [2,21,23,27,28,30,41] which referred the rate constants of these reactions to similar reactions of butanol isomers or larger alcohols. Our previous work

[42,65–68] calculated the pressure-dependent rate constants of unimolecular decomposition reactions of butanol isomers using the variable reaction coordinate-transition state theory (VRC-TST) and RRKM theory. Considering the structural similarity, the rate constants of H₂O elimination reactions of *n*-propanol (R1) and *i*-propanol (R2) in the present model are referred to those of *n*-butanol [42] and *t*-butanol [68], respectively, with the pre-exponential factor calibrated by the different degeneracies. For the C–C bond dissociation reactions, Man et al. [21] concluded from the calculation results of El-Nahas et al. [69] that the bond dissociation energies (BDEs) of C_α–C_β and C_β–C_γ bonds of *n*-propanol are almost identical to those of *s*-butanol, while the BDE of C_α–C_β bond of *i*-propanol is close to that in *n*-butanol. Thus in the present model, rate constants of the dissociation reactions of C_α–C_β (R3) and C_β–C_γ (R4) bonds of *n*-propanol and C_α–C_β bond of *i*-propanol (R5) are referred to the calculated rate constants of similar reactions of *n*-butanol [42] and *s*-butanol [66], respectively. The O–H, C–O and C–H bond dissociation reactions of propanol isomers are also included in the present model, while their rate constants are referred to similar reactions of butanol isomers or taken from the model of Sarathy et al. [2].

3.2. H-abstraction reactions and fuel radical decompositions

There are four H-abstraction reactions for *n*-propanol, producing C₃H₆OH-1 (R6), C₃H₆OH-2 (R7), C₃H₆OH-3 (R8) and NC₃H₇O (R9) radicals. For *i*-propanol, there are three H-abstraction reactions, producing IC₃H₆OH (R10), TC₃H₆OH (R11) and IC₃H₇O (R12). Here, R denotes the radicals or molecules that can attack the fuel molecules to abstract an H atom, such as H, O, OH, CH₃, HO₂ and O₂. Among the H-abstraction reactions of propanol isomers, reactions with OH attack are the only reaction family that have been extensively investigated experimentally [70–80] or theoretically [81,82]. Similar to the unimolecular decomposition reactions, the analogy method was adopted to obtain rate constants for H-abstraction reactions in most previous models of propanol isomers [2,21,23,27,28,30]. The feasibility of the analogy method to butanol isomers was further verified by Sarathy et al. [2] when compared with the measured rate constants of reactions between OH and *i*-propanol. Thus most H-abstraction reactions of *n*-propanol and *i*-propanol in the present model are referred to similar reactions of butanol isomers from the models of Cai et al. [42,66] and Sarathy et al. [83]. That is, rate constants of H-abstraction reactions at α, β, γ and hydroxyl sites of *n*-propanol are referred to those at α, β, δ and hydroxyl sites of *n*-butanol respectively, while those at α, β and hydroxyl sites of *i*-propanol are referred to those at α, β (at the methyl side) and hydroxyl sites of *s*-butanol, respectively. The H-abstraction reactions of *i*-propanol by H and CH₃ are taken from the model of Frassoldati et al. [41]. Because the lumping strategy were used for H-abstraction reactions at hydroxyl sites in their model, the rate constants in the present model are referred to similar reactions of *s*-butanol from the model of Cai et al. [66].

The most important decomposition pathways of the fuel radicals produced from the H-abstraction reactions (R6–R12) are mainly β-scission reactions. The pressure-dependent rate constants of these reactions are referred to the similar reactions of corresponding radicals of *n*-butanol and *s*-butanol. For the isomerization reactions of these fuel radicals, only the one with a five-membered ring transition state is considered, i.e. the isomerization from C₃H₆OH-3 to NC₃H₇O. The pressure-dependent rate constant is also referred to the similar reaction of the *n*-butanol radical at the C_γ site from Cai et al. [42].



4. Results and discussion

In this section, the experimental and kinetic modeling results of flow reactor pyrolysis and LBVs are discussed. Besides present experimental results, the present model is also validated against the literature data from intermediate to high temperatures including species profiles in shock tube pyrolysis, jet-stirred reactor and flow reactor oxidation, laminar premixed flame and counterflow diffusion flame, as well as global combustion parameters including ignition delay times and LBVs. The experimental targets used to validate the present model are listed in Table S1 of the *Supplementary materials*, while the comparison of the literature data and simulated results of the present model are shown in the Figs. S1–S21 of the *Supplementary materials*.

4.1. Flow reactor pyrolysis

In the flow reactor pyrolysis experiments, a series of pyrolysis products are detected and their mole fractions are derived as the function of temperature. The simulations are performed using the Plug Flow Reactor module in the Chemkin-PRO software [84]. Figures 2 and 3 show the comparisons between measured and simulated mole fraction profiles of fuels and pyrolysis products at all investigated pressures in the pyrolysis of *n*-propanol and *i*-propanol, respectively. It can be found that both hydrocarbon and oxygenated products are produced, as well as products involved in the molecular growth process, such as 1-butene (1-C₄H₈) and benzene (C₆H₆). Figures 2 and 3 also show that the present model can reasonably capture the consumption of fuels and the formation of most of the pyrolysis products.

Furthermore, the simulated results of fuels and major pyrolysis products by several previous kinetic models of propanol isomers reported by Johnson et al. [23] (referred as the Johnson model), Frassoldati et al. [41] (referred as the Frassoldati model), Man et al. [21] (referred as the Man model) and Sarathy et al. [2] (referred as the Sarathy model) are also compared at 0.04 and 1 atm, as shown in Fig. S22 of the *Supplementary materials*. It is noticed that in the *n*-propanol pyrolysis, among the four previous models, the Man model and the Sarathy model have more satisfactory performance at 1 atm, while the Frassoldati model can better capture the fuel profile at 0.04 atm. In the *i*-propanol pyrolysis, the previous models all have under-predicted the reactivity at both pressures, while the Frassoldati model generally behaves better than other three previous models. Based on the present model, modeling analysis including the rate of production (ROP) analysis and the sensitivity analysis were conducted at 0.04 atm, 1233 K and 1 atm, 1082 K in the *n*-propanol pyrolysis and at 0.04 atm, 1208 K and 1 atm, 1056 K in the *i*-propanol pyrolysis. Most of the pyrolysis products are produced abundantly under the analyzed conditions. The sensitivity analysis here is the relative response of mole fractions of the target species by changing the pre-exponential A-factor of target reaction infinitesimally.

4.1.1. Primary decomposition of fuels

Figures 2(a) and 3(a) show the results of *n*-propanol and *i*-propanol in the pyrolysis of corresponding fuels. Based on the ROP

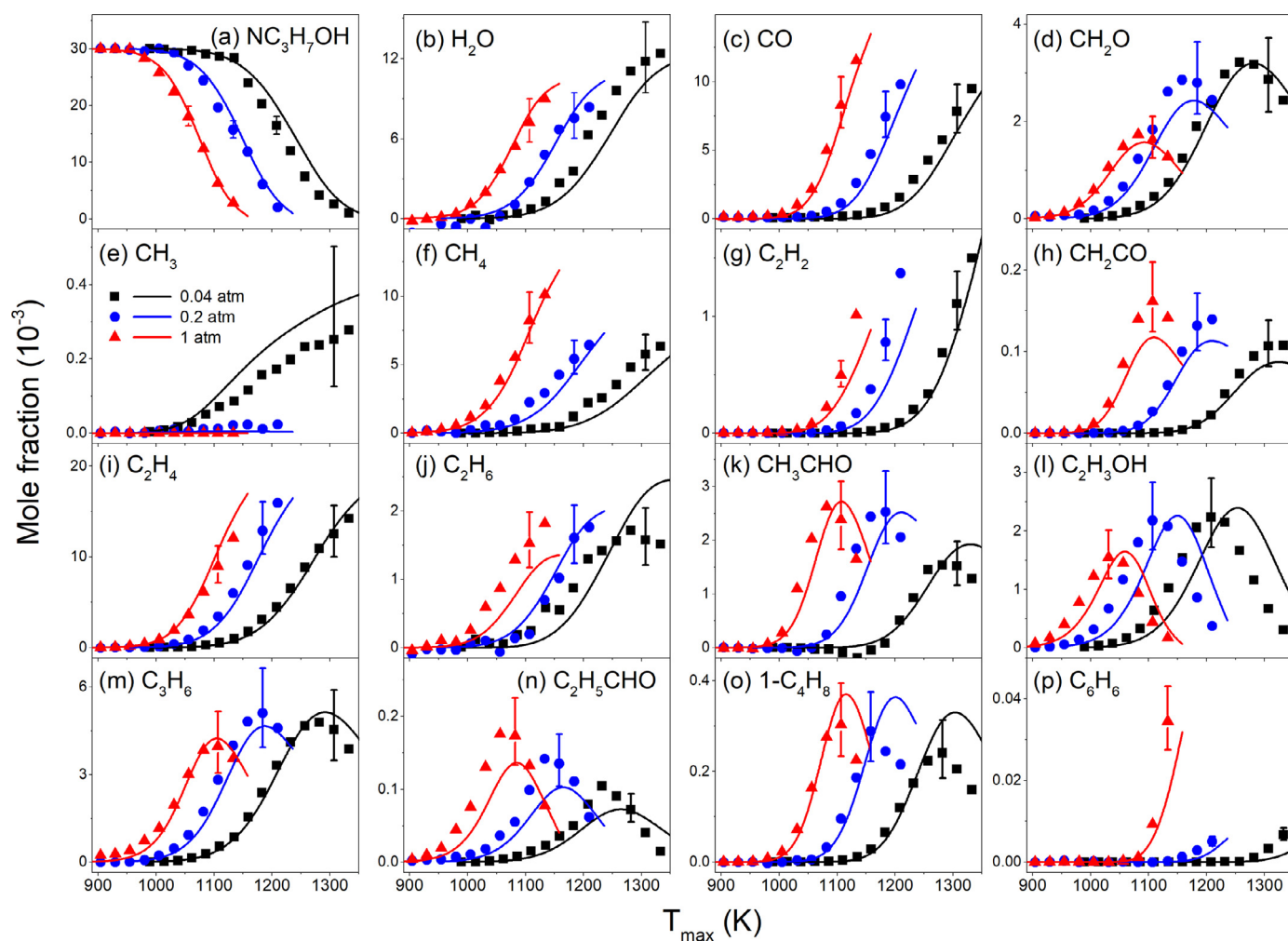


Fig. 2. Experimental (symbols) and simulated (lines) mole fraction profiles of *n*-propanol and products in the *n*-propanol pyrolysis.

analysis, the reaction networks in the pyrolysis of *n*-propanol and *i*-propanol are shown in Fig. 4. It is revealed that propanol isomers are both mainly consumed by the H₂O elimination reaction, the C–C bond dissociation reactions and the H-abstraction reactions. In general, the contributions of unimolecular decomposition reactions to the fuel consumption at 0.04 atm are higher than those at 1 atm, while the H-abstraction reactions have a reverse trend. The contributions of different reaction classes are illustrated in Fig. 5, indicating dramatic fuel isomeric effects. It can be observed that H-abstraction reactions dominates the fuel consumption in the *n*-propanol pyrolysis (Fig. 5a). In the *i*-propanol pyrolysis (Fig. 5b), the H₂O-elimination reaction (R2) has much higher contributions to the fuel consumption compared with the situation in the *n*-propanol pyrolysis, while the C–C bond dissociation reaction (R5) has lower contributions. The sensitivity analysis in Fig. 6 also reveals the dramatically enhanced importance of the H₂O-elimination reaction in the *i*-propanol pyrolysis. It also shows that the C_α–C_β bond dissociation reactions (R3 and R5), even only with very low reaction fluxes as can be seen from Fig. 5, are very crucial in determining the fuel decomposition reactivities of propanol isomers. For example, R3 has the largest negative sensitivity coefficient for *n*-propanol, while R5 has very close sensitivity coefficient for *i*-propanol compared with the H₂O-elimination reaction. Generally speaking, the unimolecular decomposition reactions with strong pressure dependence are the most sensitive reactions for fuel decomposition under the pyrolysis circumstance, which is in accordance with our previous butanol work [42].

For the effect of fuel carbon skeleton length, Wang et al. [85] compared the concentration levels of H₂O elimination products in the pyrolysis of *n*-pentanol and *n*-butanol and draw a conclusion that contributions of this reaction to normal alcohol decreased with the increasing fuel carbon skeleton length under pyrolytic conditions. The present work shows that under similar conditions, the ratio of mole fractions between propene (C₃H₆) and *n*-propanol in this work, 1-butene and *n*-butanol in the *n*-butanol pyrolysis [42], 1-pentene and *n*-pentanol in the *n*-pentanol pyrolysis [85] drops dramatically with the increasing fuel carbon skeleton length, as shown in Fig. 7. In the *i*-propanol pyrolysis, the contribution of the H₂O-elimination reaction to fuel consumption is 43% at atmospheric pressure, which is much higher than those in the *n*-butanol and *i*-butanol pyrolysis [42,65] (10% and 3% under similar condition) and are comparable to those in the *s*- and *t*-butanol pyrolysis [66,68] (36% and 66% under similar condition). This phenomenon indicates that *i*-propanol behaves more similarly to *s*-butanol and *t*-butanol in H₂O elimination, which mainly results from the more numbers of available β-H atoms for H₂O elimination in *i*-propanol, *s*-butanol and *t*-butanol than those in *n*-butanol and *i*-butanol as well as the similar position of OH moiety among *i*-propanol, *s*-butanol and *t*-butanol.

As mentioned above, H-abstraction reactions, especially those by H, OH and CH₃, play a significant role in the consumption of propanol isomers. According to the ROP analysis, those by H contribute most to the *n*-propanol consumption at 0.04 and 1 atm, followed by OH and CH₃. In the *i*-propanol pyrolysis under

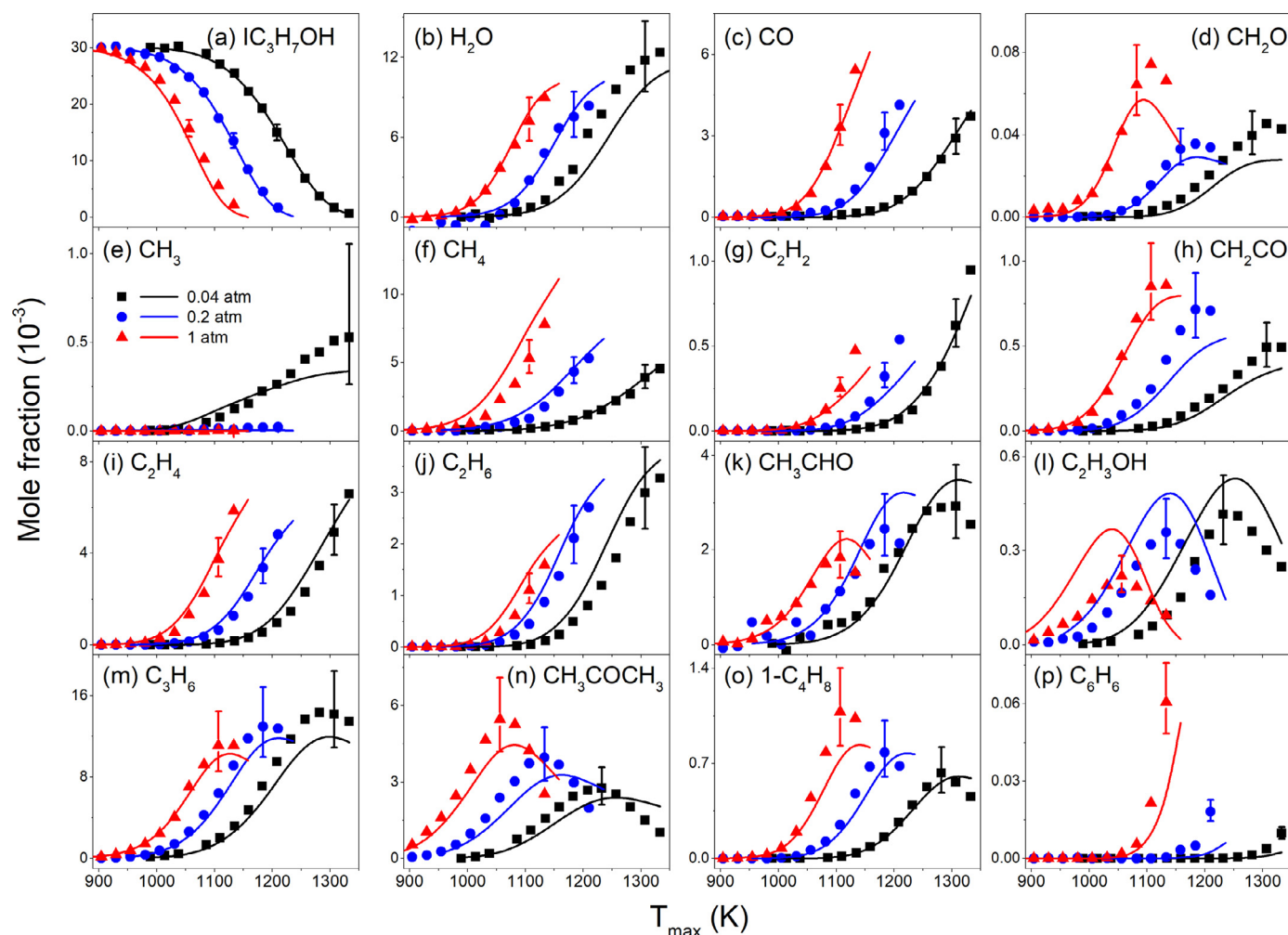


Fig. 3. Experimental (symbols) and simulated (lines) mole fraction profiles of *i*-propanol and products in the *i*-propanol pyrolysis.

atmospheric pressure, the order of contributions of H-abstraction reactions becomes H, CH₃ and OH. The more significant role of CH₃ can be explained by the doubled CH₃ groups of *i*-propanol compared with *n*-propanol. Figure 5 also compares the contributions of H-abstraction reactions at different sites. It is obvious that the contribution follows the order of $C_{\alpha} > C_{\beta} > C_{\gamma} > O$, which is in good agreement with the order of BDE [69,86]. From the sensitivity analysis in Fig. 6(a), it can also be concluded that the H-abstraction reactions at different sites play a different role in the *n*-propanol consumption. This mainly results from the ability to produce reactive radicals from the fuel radicals. As shown in Fig. 8(a), the decomposition of C₃H₆OH-1 radical produces less reactive CH₃ radical, while other fuel radicals, i.e. C₃H₆OH-2, C₃H₆OH-3 and NC₃H₇O, have the ability to release H atom and OH radical which can promote the reactivity of the whole reaction system. As a result, the H-abstraction reactions of *n*-propanol by H and OH producing C₃H₆OH-1 radical have positive sensitivity coefficients for *n*-propanol since the two reactions eventually convert H and OH to CH₃. In the *i*-propanol pyrolysis, Fig. 8(b) shows that TC₃H₆OH and IC₃H₆OH can easily decompose to H and OH radicals respectively, thus H-abstraction reactions at the two carbon sites both have negative sensitivity coefficients as shown in Fig. 6(b).

4.1.2. Formation and consumption of hydrocarbon products

Besides the differences in fuel decomposition pathways, a more pronounced fuel isomeric effect can be observed on the types and concentrations of pyrolysis products. Among hydrocarbon products,

the differences can be found not only in the decomposition products, but also in the recombination products. From Figs. 2 and 3, it can be concluded that ethylene (C₂H₄) and propene are the most abundant hydrocarbon products in the pyrolysis of *n*-propanol and *i*-propanol, respectively. In the *i*-propanol pyrolysis, propene can be produced from both the H₂O elimination of fuel (R2) and the β -C-C scission of IC₃H₆OH radical. The former reaction contributes to more than 70% of propene production at both 0.04 and 1 atm. In the *n*-propanol pyrolysis, the most important formation pathway of ethylene is the β -C-H scission reaction of ethyl (C₂H₅, R13), with a contribution of 48% at 0.04 atm and 41% at 1 atm. Ethyl is mainly produced from C_{α} -C β bond dissociation of *n*-propanol (R3) and β -C-C scission of NC₃H₇O radical. Another main formation pathway of ethylene is the β -C-C scission of C₃H₆OH-3 radical, with a contribution of 36% at 0.04 atm and 39% at 1 atm due to the enhanced role of fuel radicals at 1 atm as mentioned above. Besides, the reaction between propene and H atom (R14) also contributes about 10% to ethylene formation at both pressures.



Besides C₃ and smaller products, products with more carbon atoms than the fuels are also observed in the pyrolysis of propanol isomers, such as 1-butene and benzene. For 1-butene, the peak mole fraction in the *i*-propanol pyrolysis is 2–3 times higher than

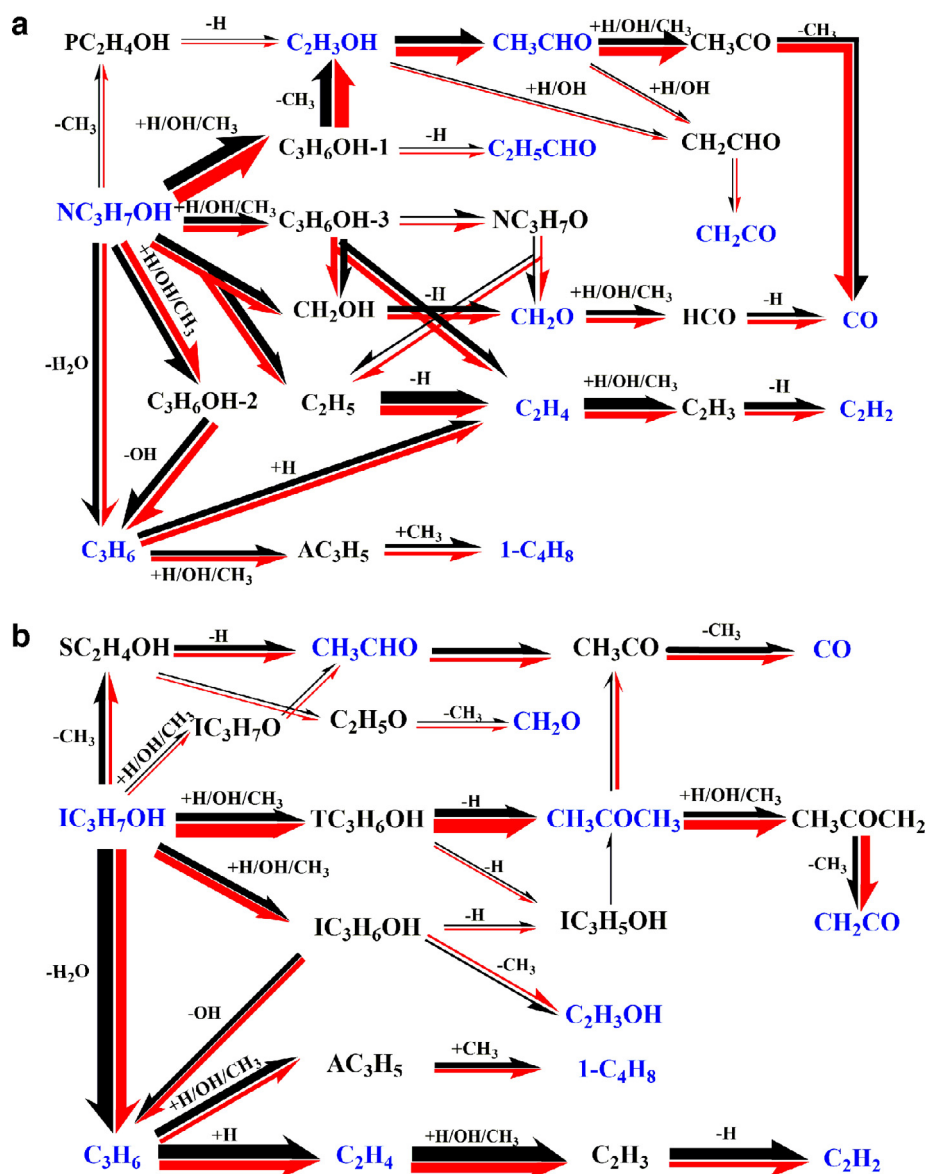


Fig. 4. Reaction networks of (a) *n*-propanol at 0.04 atm, 1233 K (black) and 1 atm, 1082 K (red) and (b) *i*-propanol at 0.04 atm, 1208 K (black) and 1 atm, 1056 K (red). The species detected in this work are marked as blue and the thicknesses of arrows represent corresponding carbon fluxes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

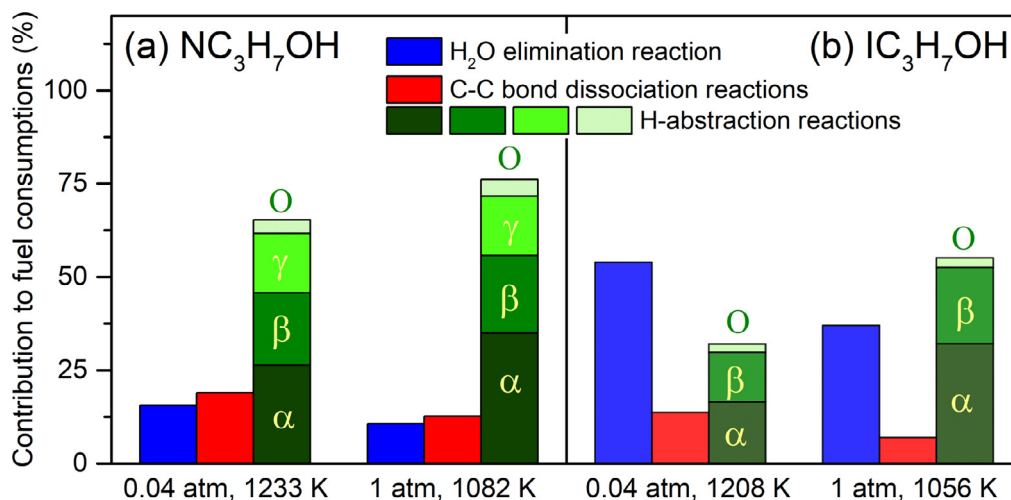


Fig. 5. Contribution of primary decomposition reactions to fuel consumption in the (a) *n*-propanol and (b) *i*-propanol pyrolysis.

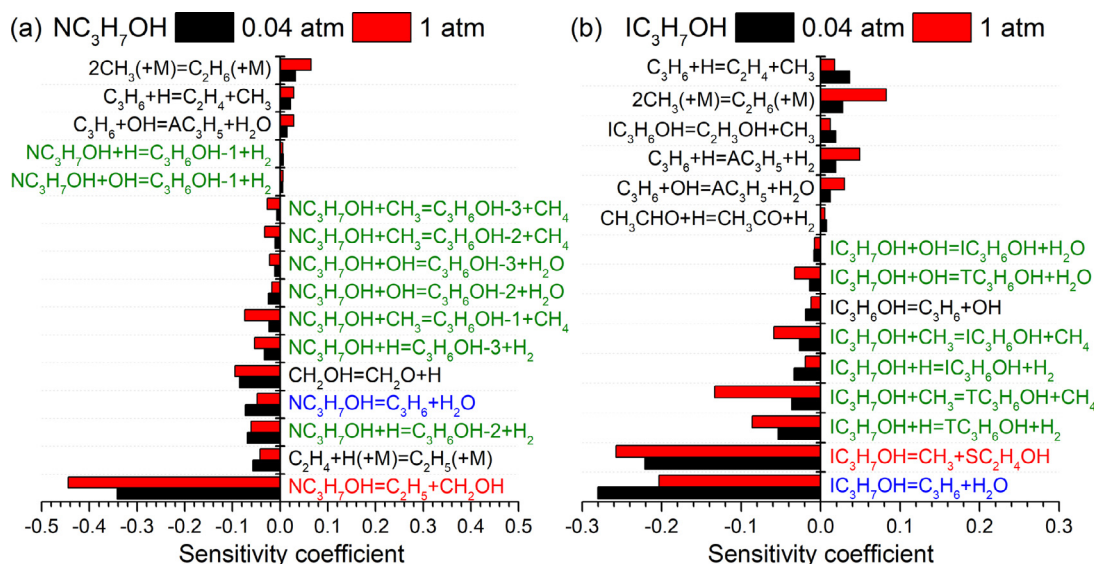


Fig. 6. Sensitivity analysis of (a) *n*-propanol at 0.04 atm, 1233 K and 1 atm, 1082 K and (b) *i*-propanol at 0.04 atm, 1208 K and 1 atm, 1056 K. The reactions marked as blue, red and green denote the H₂O elimination reactions, unimolecular decomposition reactions and H-abstraction reactions, respectively. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

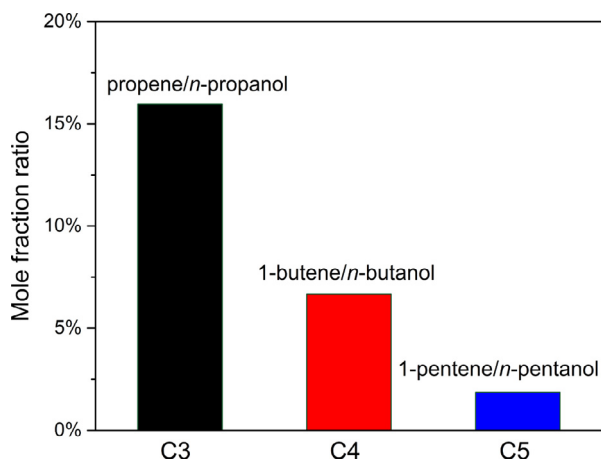


Fig. 7. The ratio of peak mole fraction of H₂O elimination product (alkenes) and the inlet mole fraction of fuel in the pyrolysis of *n*-propanol (C3 normal alcohol), *n*-butanol (C4 normal alcohol) [42] and *n*-pentanol (C5 normal alcohol) [85] at 0.04 atm.

that in the *n*-propanol pyrolysis. In the pyrolysis of both propanol isomers, the combination of allyl (AC₃H₅) and CH₃ (R15) is the dominant formation pathway of 1-butene. As shown in Figs. 2(e) and 3(e), CH₃ is more abundantly produced in the *i*-propanol pyrolysis, while allyl radical can also be more abundantly produced from the decomposition of propene in the *i*-propanol pyrolysis. Thus the much higher concentration levels of 1-butene in the *i*-propanol pyrolysis result from the abundantly produced CH₃ and allyl from the fuel decomposition pathways. Compared with 1-butene, benzene is only weakly produced in the pyrolysis of both propanol isomers, while slightly higher concentration levels of benzene can be observed in the *i*-propanol pyrolysis from Figs. 2(p) and 3(p). The formation of benzene is mainly attributed to the C₃+C₃ pathways, especially the self-combination of propargyl. The enhanced production of C₃ species from the fuel decomposition pathways explains the higher concentration levels of benzene in the *i*-propanol pyrolysis.



4.1.3. Formation and consumption of oxygenated products

The decomposition of propanol isomers can also produce many oxygenated products. Figures 2 and 3 also display the experimental and simulated mole fraction profiles of several major oxygenated species. Special attentions are paid on carbonyl products, which are major oxygenated pollutants inevitable produced in alcohol combustion. Due to the different fuel structures, differences can also be perceived among the C₁–C₃ carbonyl products between the pyrolysis of propanol isomers. In the pyrolysis of *n*-propanol, formaldehyde (CH₂O) is abundantly produced as shown in Fig. 2(d), which is in accordance with the pyrolysis of *n*-butanol [42] and *n*-pentanol [85]. This can be attributed to the structural features of normal alcohols, such as the location of CH₂OH moiety on the end of carbon skeleton and the low BDEs of C_α–C_β bond, making normal alcohols readily release the CH₂OH moiety through the C_α–C_β bond dissociation of fuels and the β–C–C scission of C_γ-site fuel radicals. In contrast, formaldehyde is very weakly produced in the *i*-propanol pyrolysis as observed from Fig. 3(d) and its formation pathway can be found in Fig. 4(b).

For C₂ oxygenated products, acetaldehyde (CH₃CHO) is abundantly produced in the pyrolysis of both propanol isomers. The ROP analysis in Fig. 4 indicates that acetaldehyde comes from different types of reaction sequences in the pyrolysis of *n*-propanol and *i*-propanol. In the *n*-propanol pyrolysis, its formation can be originated from the H-abstraction of the fuel, i.e. NC₃H₇OH → C₃H₆OH-1 → C₂H₃OH → CH₃CHO, which is similar to the case of large normal alcohols [42,85]. Thus ethenol (C₂H₃OH) becomes the most important precursor of acetaldehyde through the isomerization reaction, which can be verified by the earlier production temperatures and comparative concentration levels of ethenol as shown in Fig. 2(l). In the *i*-propanol pyrolysis, acetaldehyde is mainly produced from the reaction sequence from the unimolecular decomposition of the fuel, i.e. IC₃H₇OH → SC₂H₄OH-1 → CH₃CHO. Different from the case in the *n*-propanol pyrolysis, ethenol is much less produced in the *i*-propanol pyrolysis due to the lack of efficient formation pathways. Furthermore, different pressure effects have been observed on the mole fraction profiles of aldehyde products in the pyrolysis of propanol isomers. In the pyrolysis of *n*-propanol, Fig. 2(d,k) shows that among the three investigated pressures, formaldehyde has the highest peak mole fraction at 0.04 atm and acetaldehyde has the highest value at 1 atm,

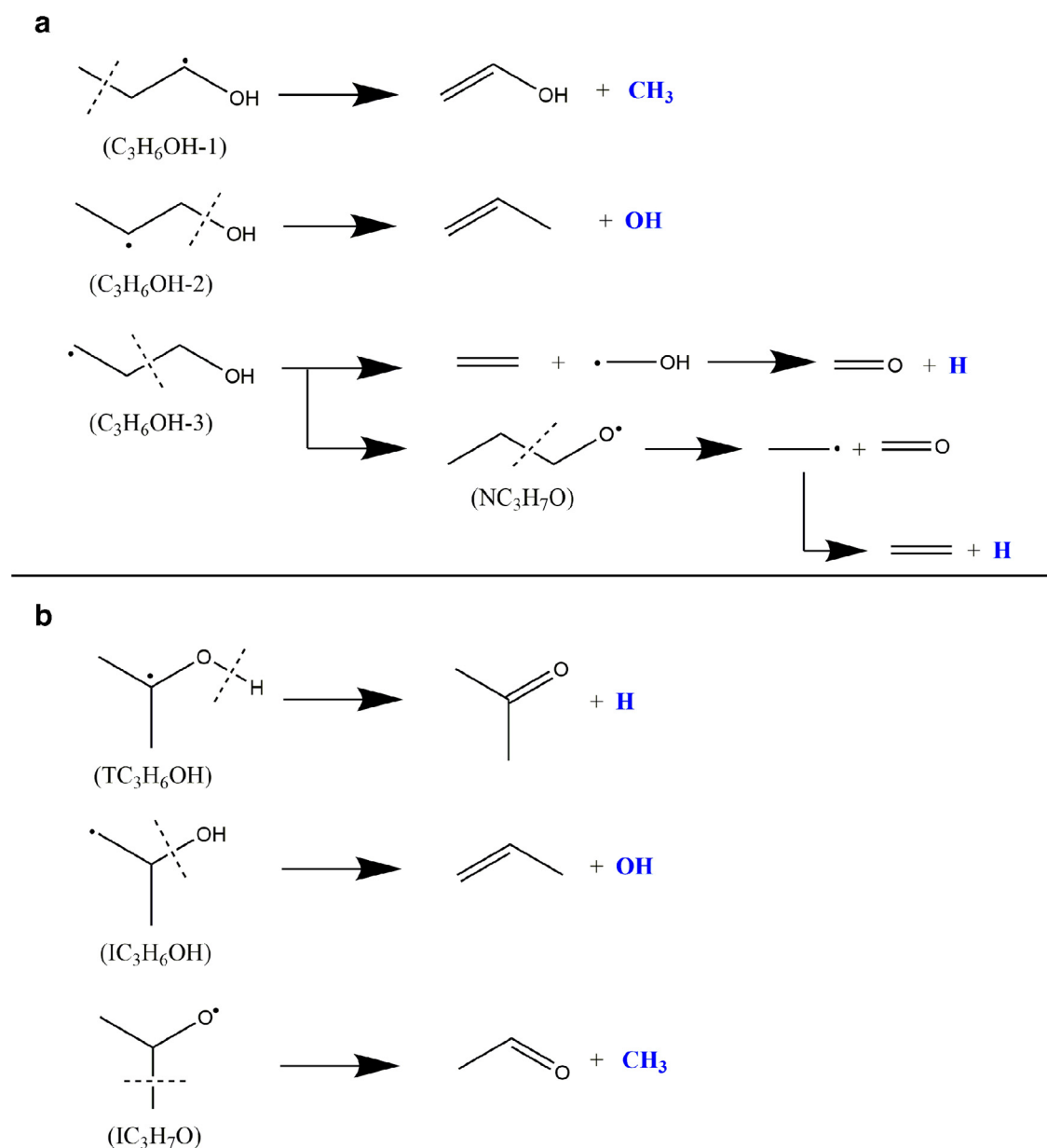


Fig. 8. Main decomposition reaction schemes of fuel radicals of (a) *n*-propanol and (b) *i*-propanol.

since formaldehyde and acetaldehyde are major products from the unimolecular decomposition and H-abstraction of *n*-propanol, respectively. In the *i*-propanol pyrolysis, Fig. 3(k) shows that the peak mole fraction of acetaldehyde has the opposite trend to that in the *n*-propanol pyrolysis, since acetaldehyde becomes a major product from the unimolecular decomposition of *i*-propanol.

Two carbonyl products, i.e. propanal (C₂H₅CHO) and acetone (CH₃COCH₃), are the dominant observed C₃ oxygenated products in the pyrolysis of *n*-propanol and *i*-propanol, respectively. The ROP analysis in Fig. 4 indicates that both of them mainly come from the decomposition of fuel radicals. Propanal is a specific C₃ oxygenated product in the *n*-propanol pyrolysis [87] via the β-C–H scission of C₃H₆OH-1 radical, while acetone is formed from the β-C–H scission of TC₃H₆OH radical in the *i*-propanol pyrolysis. However, the β-C–H scission is only a minor decomposition pathway of C₃H₆OH-1 radical due to the competition with the more favorable β-C–C scission pathway, while the β-C–H scission of TC₃H₆OH radical has no strong competition pathway. Therefore as shown in

Figs. 2(n) and 3(n), the peak mole fractions of propanal in the *n*-propanol pyrolysis are more than 20 times lower than those of acetone in the *i*-propanol pyrolysis. The feature of *i*-propanol in producing C₃ carbonyl product also influences the formation of ketene (CH₂CO). The great carbon flux from acetone through CH₃COCH₂ radical ensure the higher concentration levels of ketene in the *i*-propanol pyrolysis, as shown in Figs. 2(h) and 3(h).

Figure 9 compares the measured peak mole fractions of both aromatic pollutant (benzene) and oxygenated pollutants (mainly carbonyl pollutants) in the pyrolysis of *n*-propanol and *i*-propanol at 1 atm. It is obvious that aromatic pollutants are far less abundantly produced than oxygenated pollutants in the pyrolysis of propanol isomers. This supports the increasing concerns on the oxygenated pollutant emissions from bio-alcohols [2,88]. Among the two fuel isomers, *i*-propanol yields higher pollutant content than *n*-propanol, both for aromatic and oxygenated pollutants, which adds more evidences for the strong influence of fuel molecular structures on pollutant emissions in alcohol combustion.

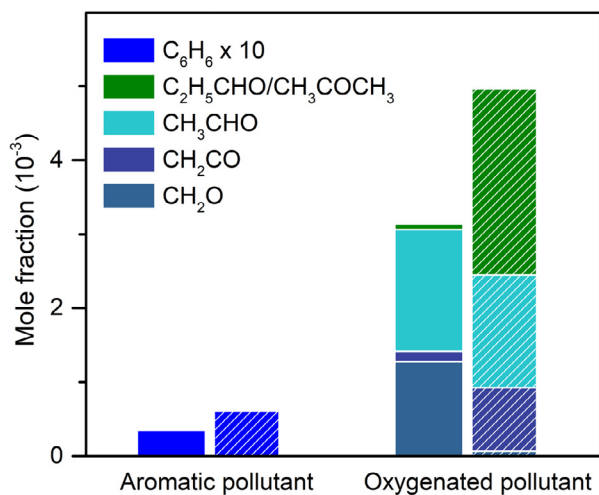


Fig. 9. Comparison between measured concentration levels of aromatic and oxygenated pollutants in the pyrolysis of *n*-propanol (solid columns) and *i*-propanol (slash columns) at 1 atm.

4.2. Laminar burning velocities

4.2.1. Experimental and simulated results

Figure 10 shows the measured and simulated LBVs of propanol isomers at 1–10 atm and 423 K in this work. Due to the influence of wrinkled flame surface caused by cellular instabilities under large equivalence ratio conditions at elevated pressures, the available equivalence ratio range for LBV measurements are limited up to 1.3 and 0.9 at 5 and 10 atm, respectively. The simulations are performed using the Premixed Laminar Flame Speed Calculation module of Chemkin-PRO software [84] with the consideration of the Soret effect. A domain of 10 cm and a maximum grid points of 1000 (GRAD = 0.1, CURV = 0.1) are used to provide sufficient refinement. It can be observed that the present model can well capture the new LBV data in this work. Furthermore, the present model can also reasonably predict measured LBVs of propanol isomers in literature [25–27,29–32], as shown in Figs. S20–S21 of the Supplementary materials.

It can be seen from Fig. 10 that for both propanol isomers, the LBV results at specific pressures, e.g. 1 and 3 atm where whole profiles can be obtained, peak at ϕ around 1.1, while the results decline with increasing pressure as expected. As seen from Fig. 10, *n*-propanol has higher LBVs than *i*-propanol at all investigated pressures, which is consistent with observations of previous investigations [27,29,89]. In this work, the difference in the measured LBV values can reach up to 8.8 cm/s at 1 atm. This indicates that *n*-propanol is much more reactive than *i*-propanol under flame conditions, which again reveals obvious fuel isomeric effects. Furthermore, the higher reactivity of *n*-propanol found in the LBV results is in good agreement with previous findings in ignition delay times, that is, *n*-propanol shows generally faster ignition delay times than *i*-propanol [20,21,23]. Similar phenomena that the existence of branched structure tends to decrease the reactivity compared to normal structure have also been found in hydrocarbon combustion [44,90].

Figure 11 shows the measured Markstein lengths in this work. For both *n*-propanol and *i*-propanol, an overall decrease trend of Markstein length is observed with the increase of equivalence ratio under all investigated pressures. At a specific equivalence ratio, it is noticed that the Markstein length decreases with the increase of pressure at $\phi < 1.4$, while a reverse trend happens at $\phi = 1.5$ and pressures of 1 and 3 atm. The transition from positive to negative Markstein length are observed around $\phi = 1.3$ at 1–5 atm for both propanol isomers. In general, *n*-propanol and *i*-propanol flames show similar behaviors in Markstein length and in turn close instability characteristics.

4.2.2. Kinetic effects on LBVs

It is widely accepted that for a specific fuel, the LBV values are strongly dependent on adiabatic flame temperature and fuel-specific kinetics. In terms of the propanol isomers, the difference in their adiabatic flame temperatures is so small that the influence of adiabatic flame temperature is often considered to be negligible [27]. As a result, the kinetic effects become significant in the laminar flame propagation of propanol isomers. In order to provide more insight into the kinetic effects, the sensitivity analysis and ROP analysis are conducted for propanol flames. Figure 12 shows the sensitivity analysis results of LBVs of lean ($\phi = 0.6$) and rich ($\phi = 1.5$) flames of propanol isomers at 1 and 10 atm. The sensitivity analysis here is the relative response of

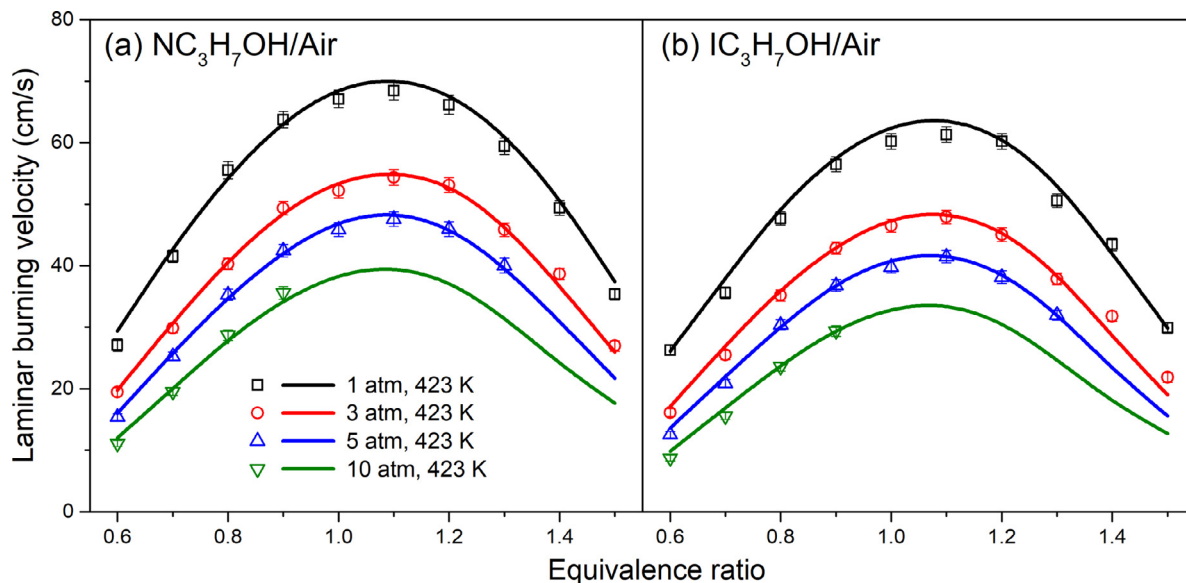


Fig. 10. Experimental (symbols) and simulated (lines) LBVs of (a) *n*-propanol and (b) *i*-propanol at 1–10 atm and 423 K.

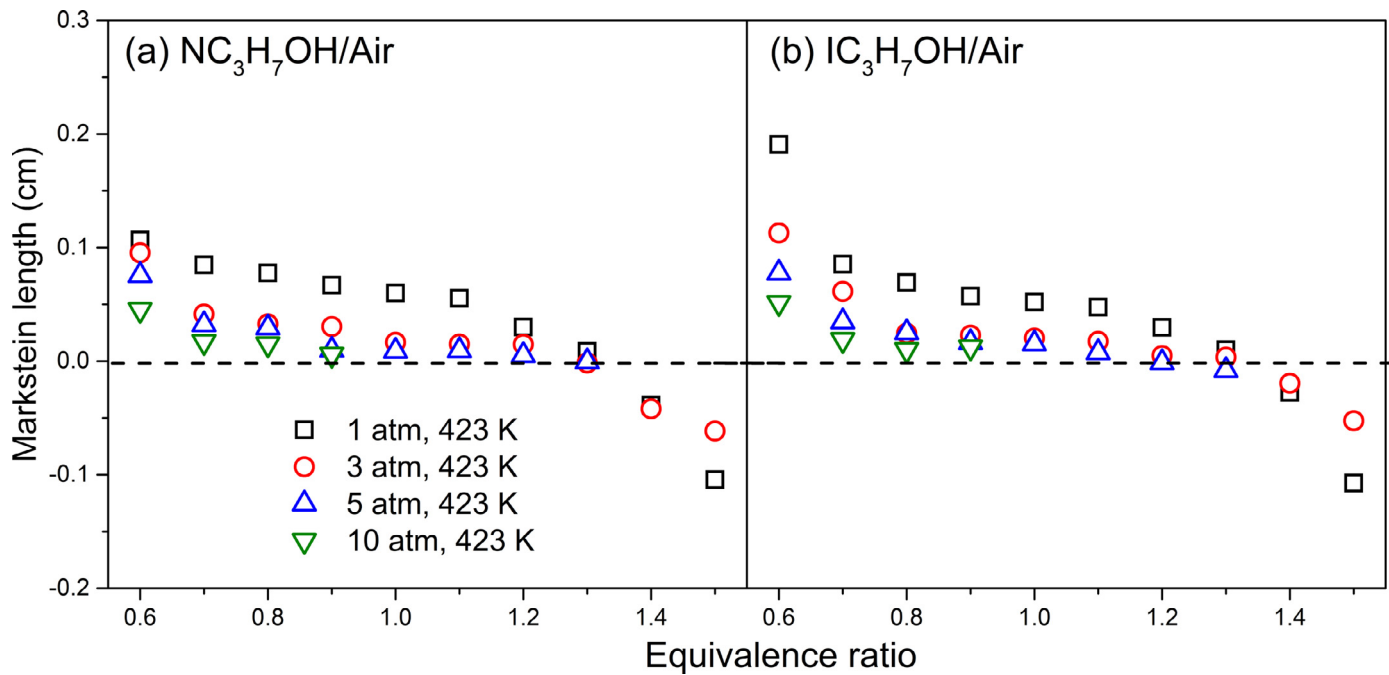


Fig. 11. Measured Markstein lengths of (a) *n*-propanol and (b) *i*-propanol at 1–10 atm and 423 K.

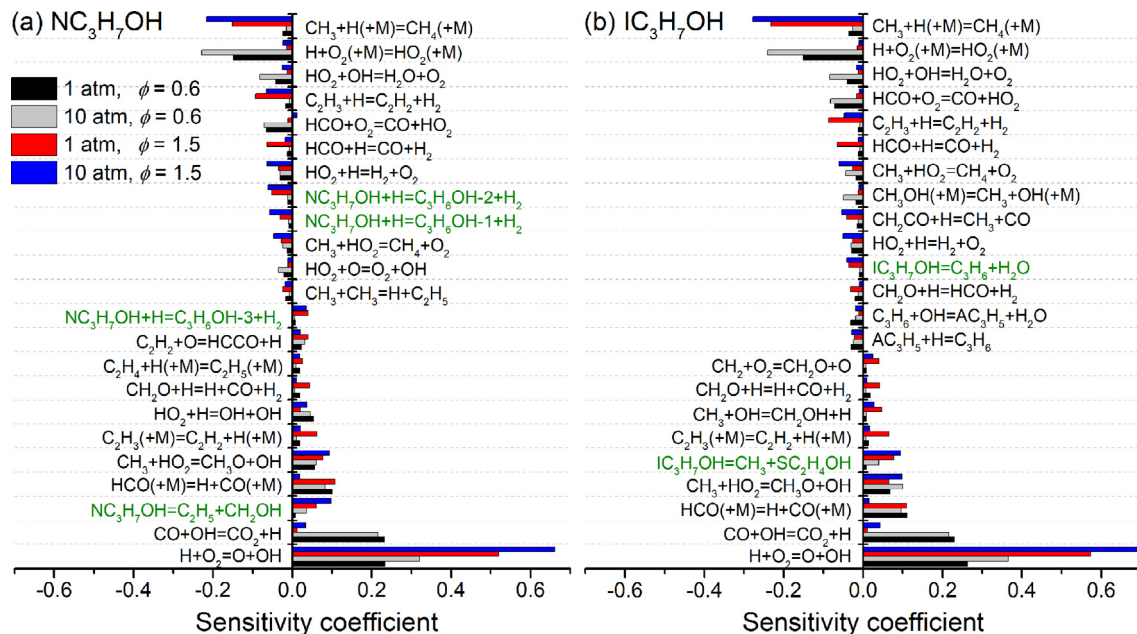


Fig. 12. Sensitivity analysis of LBVs of (a) *n*-propanol and (b) *i*-propanol under different equivalence ratio and pressure conditions. Green color denotes fuel reactions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

LBV by changing the pre-exponential A-factor of target reaction infinitesimally. It can be observed that the dominant reactions for the laminar flame propagation of propanol isomers are quite similar. The most important high-temperature chain-branching reaction $\text{H} + \text{O}_2 = \text{O} + \text{OH}$ has the highest positive sensitivity coefficients, which is followed by the key chain-propagation reaction $\text{CO} + \text{OH} = \text{CO}_2 + \text{H}$. Unimolecular decomposition reactions of formyl (HCO), vinyl (C_2H_3) and C_2H_5 radicals are also among sensitive reactions in *n*-propanol and *i*-propanol flames, since these reactions can produce H atom. The two chain-termination/inhibition

reactions consuming H atom, i.e. $\text{CH}_3 + \text{H} (+\text{M}) = \text{CH}_4 (+\text{M})$ and $\text{H} + \text{O}_2 (+\text{M}) = \text{HO}_2 (+\text{M})$, have the highest negative sensitivity coefficients. In general, the chain-termination reactions which greatly consume radicals, such as $\text{CH}_3 + \text{H} (+\text{M}) = \text{CH}_4 (+\text{M})$, $\text{CH}_3 + \text{HO}_2 = \text{CH}_4 + \text{O}_2$, $\text{CH}_3\text{OH} (+\text{M}) = \text{CH}_3 + \text{OH} (+\text{M})$ (in reverse direction) and $\text{AC}_3\text{H}_5 + \text{H} = \text{C}_3\text{H}_6$, have greater negative sensitivity coefficient in *i*-propanol flames than in *n*-propanol flames. Furthermore, key unimolecular decomposition reactions of *n*-propanol and *i*-propanol are also sensitive to laminar flame propagation, especially under rich conditions. For example, C–C bond dissociation

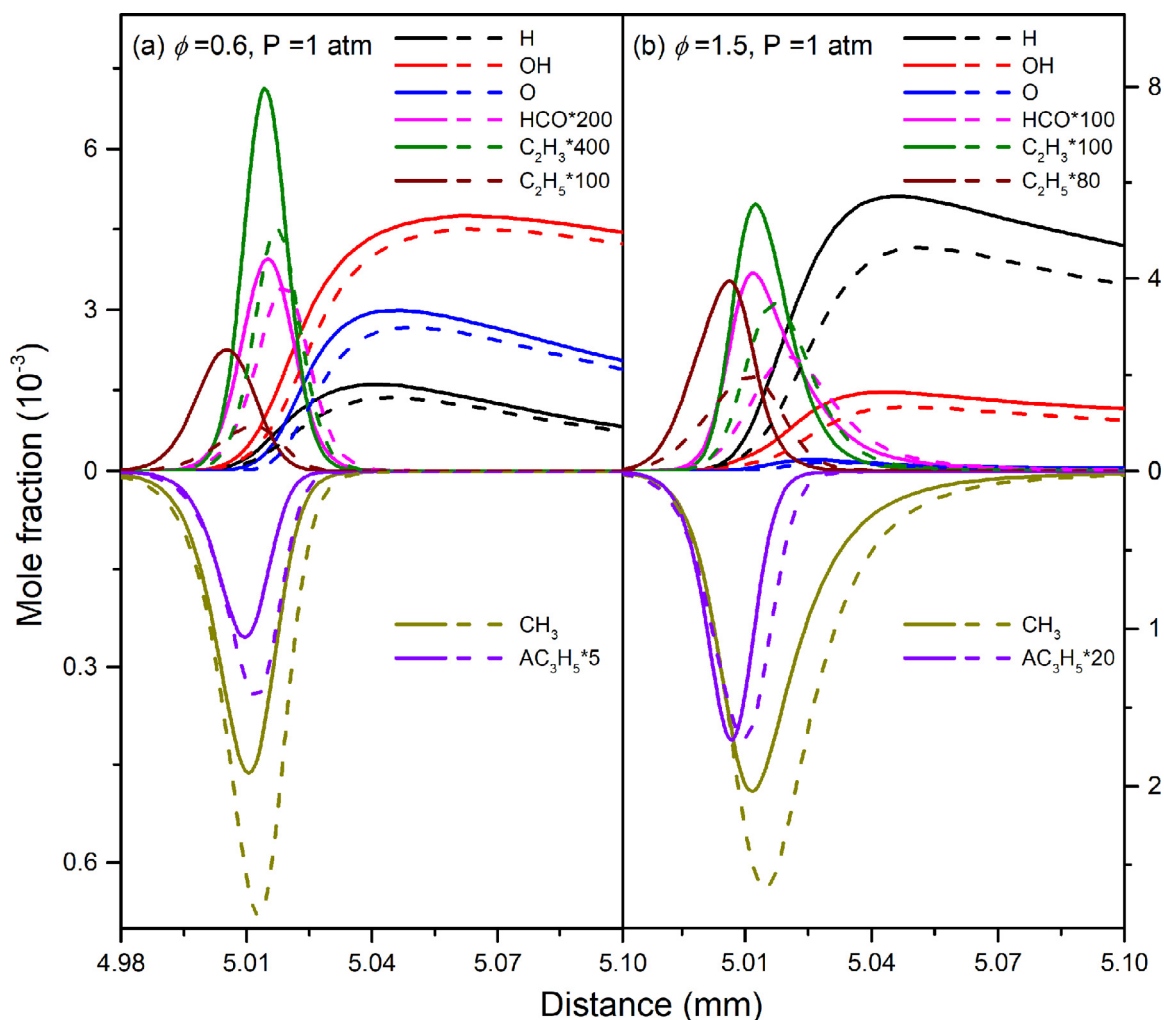


Fig. 13. Simulated mole fraction profiles of key radicals in *n*-propanol (solid lines) and *i*-propanol (dash lines) flames at 1 atm under (a) lean and (b) rich conditions. In each figure, the upper part shows radicals (H, OH, O, formyl, vinyl and ethyl) with higher mole fractions in *n*-propanol flames, while the lower part shows radicals (CH_3 and allyl) with higher mole fractions in *i*-propanol flames.

reactions of *n*-propanol (R3) and *i*-propanol (R5) both have considerable positive sensitivity coefficients, while H_2O -elimination reaction of *i*-propanol (R2) has negative sensitivity coefficients.

It is recognized that concentrations of key radicals have great influence on the laminar flame propagation [2,89]. Figure 13 compares the simulated mole fraction profiles of eight key radicals in *n*-propanol and *i*-propanol flames at 1 atm, while the comparison at 10 atm is shown in Fig. S23 in the *Supplementary materials*, which shows very similar results. To guide the eye, the profiles for radicals with higher mole fractions in *i*-propanol flames are shown in the reverse direction to those with higher mole fractions in *n*-propanol flames. In general, the shapes of radical profiles in *n*-propanol and *i*-propanol flames are very similar. In both *n*-propanol and *i*-propanol flames, H, O, OH and CH_3 have the highest mole fractions among all radicals. Among the four radicals, OH and O have higher mole fractions in lean flames, while H and CH_3 becomes more prevailing in rich flames.

As shown in Fig. 13, the peak mole fractions of radicals have apparent differences in *n*-propanol and *i*-propanol flames. It is observed that, among the eight radicals, H, OH, O, formyl, vinyl and ethyl radicals have higher peak mole fractions in *n*-propanol flames under both lean and rich conditions, while CH_3 and allyl radicals have higher peak mole fractions in *i*-propanol flames. According to the ROP analysis, more than 80% of the fuel is consumed by H-abstraction reactions (R6–R9) to form four fuel

radicals in *n*-propanol flames, while among the rest reactions the $\text{C}_\alpha\text{--C}_\beta$ bond dissociation reaction producing ethyl and CH_2OH radical (R3) has the highest contributions. Almost all produced radicals mainly undergo β -scission reactions or isomerization reactions, eventually yielding abundant ethylene and formaldehyde. In *i*-propanol flames, H-abstraction reactions at the primary (R10) and tertiary (R11) carbon sites dominate the fuel consumption with the contributions of over 40% and 20%, respectively, while the $\text{C}_\alpha\text{--C}_\beta$ bond dissociation reaction (R5) and H_2O -elimination reaction (R2) consume the rest. Further decomposition reactions of the intermediates formed from these pathways are prone to produce propene and acetone rather than ethylene and formaldehyde. The above simulated product distributions are in accordance with the observations in the speciation studies of propanol flames by Li et al. [39] and Kasper et al. [38].

As expected, the fates of ethylene and formaldehyde under flame conditions are to produce vinyl and formyl, respectively. Together with the easy formation of ethyl from R3, formyl, vinyl and ethyl are more abundantly produced in *n*-propanol flames, as shown in Fig. 13. The decomposition of formyl, vinyl and ethyl radicals can all enhance the formation of H atom, which explains the higher positive sensitivity coefficients of these reactions in *n*-propanol flames than in *i*-propanol flames. In contrast, the fates of propene and acetone are prone to boost allyl and CH_3 , respectively. Together with the easy formation of CH_3 from R5, CH_3 and allyl

are more abundantly produced in *i*-propanol flames. According to above sensitivity analysis, the crucial chain-termination reactions in propanol flames are all recombination reactions of CH_3 and allyl with H, OH and HO_2 radicals. As a result, the strong fuel isomeric effects on laminar flame propagation of propanol can be explained. The linear alcohol structure of *n*-propanol facilitates the production of active radicals like formyl, vinyl and ethyl, which enhances the formation of H and consequently promotes the laminar flame propagation of *n*-propanol. The branched alcohol structure makes *i*-propanol tend to produce stable radicals like CH_3 and allyl, which reduces the concentrations of H and OH through recombination reactions and consequently inhibits the laminar flame propagation of *i*-propanol.

5. Conclusions

In this work, flow reactor pyrolysis and laminar flame propagation of propanol isomers are investigated under same conditions. Flow reactor pyrolysis measurements are performed at 0.04–1 atm using SVUV-PIMS. It is observed that hydrocarbon products mainly differ in concentration levels, while great differences in oxygenated products exist on both molecular structures and concentration levels. Ethylene and propene are most abundant hydrocarbon products in the *n*-propanol and *i*-propanol pyrolysis, respectively. The most abundant oxygenated products are formaldehyde, acetaldehyde and ethenol in the *n*-propanol pyrolysis, while acetone and acetaldehyde are the most abundant oxygenated products in the *i*-propanol pyrolysis. Aromatic pollutants are far less abundantly produced than oxygenated pollutants in the pyrolysis of propanol isomers, while *i*-propanol yields higher pollutant content than *n*-propanol, both for aromatic and oxygenated pollutants. Laminar flame propagation measurements of propanol isomers are carried out at 423 K, pressures of 1–10 atm and equivalence ratios of 0.6–1.5. A general trend that *n*-propanol has much faster LBFs than *i*-propanol is noticed under all investigated conditions.

A kinetic model of propanol isomers is developed and validated against the new data in this work. According to the modeling analysis, H-abstraction reactions, combined with the $\text{C}_\alpha\text{--C}_\beta$ bond dissociation reaction, dominate the consumption of *n*-propanol in its pyrolysis, leading to the abundant production of ethylene and formaldehyde. The H_2O -elimination reaction and H-abstraction reactions dominate the fuel consumption in the *i*-propanol pyrolysis, leading to abundant production of propene and acetone. In laminar flame propagation, different radical pool distributions of propanol isomers are illustrated and found to be largely influenced by isomeric fuel structures. The linear alcohol structure makes *n*-propanol tend to produce high concentration levels of formyl, vinyl and ethyl, which enhances the formation of H and consequently promotes the laminar flame propagation of *n*-propanol. The branched alcohol structure of *i*-propanol facilitates the production of stable radicals like CH_3 and allyl, which reduces the concentrations of H and OH through recombination reactions and consequently inhibits the laminar flame propagation of *i*-propanol. In addition, the present model also shows satisfactory predictions on the literature data, indicating the reliability of this model over a wide range of temperatures, pressures and equivalence ratio.

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Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2019.05.040.

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